

Test Report on the Municipal Financial Management System (MFMS)

1. Modules/ Files developed

Files	What it is
tests/run_tests.sh	Automated test script: 44 black-box tests that the system must pass, plus 3 tests for known bugs
Test documentation	Test plan and results, and the bug list in section 4 of this record

How the test script works;

- For each test it sends keyboard input to the program(for example a menu choice, a name and a salary), then checks that the expected text appears on screen(for example “Employee added successfully” or “Negative budget not allowed”). Tests that use the simple data start the program with -demo. Each test prints PASS or FAIL, and summary is printed out.

Test types used;

- Normal input, invalid input(negative numbers, letters, empty fields, out-of-range menu choices), Duplicate IDs, searching for records that exist and that do not exist, calculation checks, reports with and without data, and end of output.

2. Testing Performed

Git bash: I ran `bash tests/run_tests.sh` in Git bash which gave me a result: 53 passed, 0 failed, plus 3 known-bug

```
MINISW64\c:\Users\hnamu\Videos\MFMS_group\MFMS_group
C:\Users\hnamu\Videos\MFMS_group\MFMS_group
$ cd "/C:/Users/hnamu/Videos/MFMS_group/MFMS_group"
$ bash tests/run_tests.sh
--- MFMS TEST RUN ---
Main menu ---
PASS T01 Main menu shows all options
PASS T02 Main menu: letters rejected
PASS T03 Main menu: out-of-range rejected
PASS T04 Demo mode loads sample data
--- Employee management ---
PASS T05 Add employee (valid)
PASS T06 Negative basic salary rejected
PASS T07 Empty name rejected
PASS T08 Name with digits rejected
PASS T09 Duplicate employee ID rejected
PASS T10 Salary calculation correct
PASS T11 Search employee by ID
PASS T12 Search employee by partial name
PASS T13 Search employee: not found
--- Budget management (group module) ---
PASS T14 Add budget within limit
PASS T15 Add budget over limit
PASS T16 Negative budget rejected
PASS T17 Negative expenditure rejected
PASS T18 Budget menu: invalid number
PASS T19 Display budgets (demo)
PASS T20 Search budget: found
PASS T21 Search budget: not found
PASS T22 Departments exceeding budget
PASS T23 Exceeded list when none exceed
--- Supplier management (group module) ---
PASS T24 Add supplier (valid)
PASS T25 Invalid email rejected
PASS T26 Invalid phone rejected
PASS T27 Duplicate supplier ID rejected
PASS T28 Display suppliers (demo)
PASS T29 Search supplier by ID
PASS T30 Search supplier by name
PASS T31 Search supplier: not found
PASS T32 Supplier menu: invalid number
--- Asset management (group module) ---
PASS T33 Add asset (valid)
PASS T34 Negative asset value rejected
PASS T35 Asset: letters as value rejected
PASS T36 Asset: empty name rejected
PASS T37 Asset: invalid type choice
PASS T38 Duplicate asset ID rejected
PASS T39 Asset menu: invalid number
PASS T40 Display asset register (demo)
PASS T41 Search asset by ID
PASS T42 Search asset by type
PASS T43 Search asset by department
PASS T44 Search asset: not found
--- Reports ---
PASS T45 Employee report values
PASS T46 Budget report totals
PASS T47 Supplier report
PASS T48 Asset report (from reports menu)
PASS T49 Asset report (from asset menu)
PASS T50 All reports with no data (no crash)
PASS T51 Budget entered then reported
PASS T52 Supplier entered then reported
--- Robustness / known bugs in group modules ---
PASS T53 End of input at main menu exits
BUG BUG1 Budget menu: letters should be rejected (program hangs)
BUG BUG2 Budget: letters at amount prompt (program hangs)
BUG BUG3 Budget: duplicate department should be rejected
=====
Passed: 53  Failed: 0  Known bugs: 3  Bugs fixed: 0
C:\Users\hnamu\Videos\MFMS_group\MFMS_group
$
```

- The tests also cover the following areas :

Area	Examples of what was checked
Main menu	All options shown: letters and out-of-range numbers rejected
Employees	Add employees; negative salary, empty name and duplicate ID rejected; salary calculation; search by ID and by name
Budget	Remaining budget and status; negative budget and expenditure rejected; invalid menu choice; search; departments over budget
Suppliers	Add supplier; invalid email and phone rejected; display; search by ID and name; not-found case
Assets	Add assets; negative value rejected; search; display register
Reports	All four reports(employee, budget, supplier, asset) with sample data; reports with no data don not crash; reports after entering new data

Manual testing; I ran the program with ./mfms.exe –demo and went through every menu, search and report. I also tried entering invalid details to see if it will go through or not, and it didn't go through.

Compilation; the project complies with gcc -std=c99 -Wall -Wextra -pedantic without warnings

3. Bugs found

ID	Module	Problem	Steps to reproduce
BUG 1	Budget	Trying letters at the budget menu makes the program loop forever	Main menu 2 type abc

BUG 2	Budget	Trying letters at the allocated budget prompt loop forever	2 1 department name type abc
BUG 3	Budget	A department with the same name can be added twice	Add "Finance" twice
NOTE	Suppliers	Telephone check accepts digits only, so +264... is rejected	3 1 enter a phone number starting with +